FINAL PROJECT REPORT
ARKANSAS FILM DATABASE

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APRIL 30, 2026

TABLE OF CONTENTS

CLIENT INTRODUCTION/ORGANIZATION BACKGROUND 3
ANALYSIS 3
The Problem 3
Desired Outcome 3
Options and Solutions 4
Project Plan 4
DESIGN & DEVELOPMENT WORK PERFORMED 5
Project Design 5
Project Development 5
Testing and Evaluation 6
Project Implementation 6
PROFESSIONAL RESPONSIBILITIES 7
Ethics, Legality, and Security 7
SUPPORT AND INTEGRATION 7
Ongoing Training 7
Long-Term Support and Maintenance 7
SUMMARY/LESSONS LEARNED 8
Positive Outcomes 8
Lessons Learned 8
LIST OF REFERENCE SOURCES 8
APPENDICES 9
Hours Worked 9
Documentation 9

CLIENT INTRODUCTION/ORGANIZATION BACKGROUND

The Arkansas Economic Development Group (AEDG) is an organization focused on promoting economic growth, tourism, and cultural awareness within the State of Arkansas. One opportunity identified by the organization was the lack of a modern digital platform highlighting Arkansas's contributions to the film and television industry.

The organization's goal for this project was to develop a proof-of-concept website that could market Arkansas-related films, actors, actresses, and filming locations to the public. The intended users include movie enthusiasts, Arkansas residents, tourists, and businesses interested in the state's creative economy.

The AEDG required a scalable, modern solution that could be maintained easily while demonstrating the value of future investment in Arkansas film promotion initiatives. The overall budget for this project was $30,000, and the target completion date was April 30, 2026.

ANALYSIS

The Problem

The Arkansas Economic Development Groups seeks to strengthen awareness of Arkansas' film industry and improve the state's visibility as a competitive filming destination while also highlighting the actors and actresses with ties to Arkansas. Many states actively promote their film heritage through modern digital platforms. Arkansas currently lacks a primary interactive source to address this gap.

Desired Outcome

The desired outcome involves a seamless, integrated digital solution that automates the data retrieval process and provides immediate film/actor/actress information in the form of a display card through a modern web interface.

- **Data:** Structured data is retrieved from The Movie Database (TMDB) API and displayed in an easy-to-read detail card format.

- **Process:** Users can browse films and talent pages or search by film title or actor name. Search logic checks movies first, then people if no movie match is found.

- **People:** End users benefit from a simple, intuitive interface. Organizational staff benefit from a low-maintenance promotional platform powered by API-driven content.

- **Technical Improvements:** A responsive client-side web application built with HTML5, CSS, and JavaScript using reusable templates and real-time API integration.

Options and Solutions

Two main options were considered:

Option 1: Commercial Website Platform
A pre-built commercial website like WordPress would have been the quickest option. However, we considered the potential high cost of licensing and/or subscription maintenance and decided that a custom build would be more economical. We also took into account the limitations on customization when using a product built for generic use. It made more sense to design an application from scratch that we knew would work.

Option 2: Custom Website with API Integration
A custom-built solution using HTML5, CSS, JavaScript, and TMDB API integration offered greater design flexibility, lower recurring cost, and stronger demonstration of technical skills.

The team recommended option two to build the website from scratch.

Project Plan

The project was executed following the phased approach detailed in the Project Charter and was completed ahead of schedule.

Project Milestone Schedule	**Target Date**	**Actual Date**	**Status**
Project Start	02/03/2026	02/03/2026	Complete
API Research and Documentation Review	02/20/2026	02/16/2026	Complete
Template and UI Design	03/10/2026	03/08/2026	Complete
API Integration Complete	03/25/2026	03/23/2026	Complete
Film Pages Functional	04/05/2026	04/03/2026	Complete
Actor/Actress Pages Functional	04/12/2026	04/10/2026	Complete
System Testing/Debugging	04/19/2026	04/16/2026	Complete
Deploy Solution	**04/23/2026**	**04/21/2026**	**Complete**
Project Complete	**04/30/2026**	**04/28/2026**	**Complete**

- **Budget/Resource Constraints:** The project was constrained by a total effort budget of 245 hours of labor time, which was successfully met. Hardware and software were constrained to be compatible with current organization IT platforms and required no new costs, adhering to the budget.

- **Skills/Training Needs:** The project knowledge of frontend web development (HTML, CSS, JavaScript) and experience with API integration and data visualization. These skills were managed internally by the Project Manager and the IT support team.

DESIGN & DEVELOPMENT WORK PERFORMED

Project Design

The solution utilizes a lightweight client-side web application architecture with minimal backend requirements.

IT Infrastructure: Static hosting environment with browser-based execution.

- **Hardware:** Standard desktops, laptops, tablets, and mobile devices.
- **Software (Development):** Github (repository and hosting service for development), Git (version control), Notepad (code editor), and web browsers.
- **Software (Runtime):** Modern web browsers supporting HTML5, CSS, and JavaScript.
- **Environments:** Separate development/testing environment before deployment.

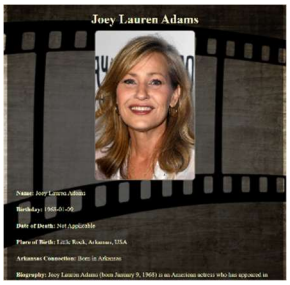
Project Development

Data Architecture: The Arkansas Film Database does not store personal user data. Information is processed temporarily in the browser using Javascript.

- **External Data:** TMDB API.
- **Returned Data:** Titles, genres, budgets, images, biographies, cast, birthdays, filmographies, and synopsis.
- **Local Supplemental Data:** Arkansas connections and awards manually mapped by TMBD IDs.
- **Website's Interface/Interactions:** The user interface was designed for simplicity, proof of concept, and ease of use. It is intuitive and functions responsively on various web browsers.

Interface/Interactions: The user interface was designed for clarity, simplicity, and intuitive navigation.

- Homepage search bar.
- Films page with clickable posters.
- Actors/Actresses page with clickable profiles.
- Dynamic film detail pages.
- Dynamic actor/actress detail pages.
- Clickable cast members and filmography links.



Code/Web Application Development: The application was developed using:

- **HTML5:** Structure and organization of the user interface.
- **CSS:** Responsive design, navigation styling, card layouts, and theme presentation to ensure the appearance and functionality appeals to the target audience.
- **JavaScript:**
  - API calls using fetch().
  - JSON processing.
  - Search logic.
  - URL parameter handling.
  - Dynamic page population.
  - Clickable relationship links between entities.
  - Error handling and fallback messages.

Testing and Evaluation

Testing focused on functionality, integration, and usability.

- Unit testing of JavaScript functions.
- API endpoint validation.
- Search testing.
- Navigation testing between pages.
- Browser compatibility testing.
- Layout responsiveness testing.
- User acceptance review.

Results: The website performed successfully and was approved for deployment.

Project Implementation

The solution was successfully **deployed** on **April 21, 2026**, using GitHub pages for hosting.

PROFESSIONAL RESPONSIBILITIES

Ethics, Legality, and Security

The development and deployment of the Arkansas Film Database adhered to the professional computing standards of professional conduct including principles consistent with the ACM Code of Ethics.

Ethical Considerations

- Honest representation of information.
- Public benefit through educational/promotional use.
- Accessible and intuitive design.
- Respect for intellectual property.

Legal/Compliance Considerations

- TMDB API terms of service were reviewed.
- Public attribution maintained where required.
- No copyright content stored locally beyond allowed use.

Security Considerations

- HTTPS API requests used.
- No user account or passwords required.
- No personal data collected.
- Minimal attack surface due to static client-side design.

SUPPORT AND INTEGRATION

Ongoing Training

Organization staff received guidance regarding:

- Site structure.
- How API-based content is delivered.
- Troubleshooting connectivity issues.
- Updating static content and links.

Long-Term Support and Maintenance

Recommended support model:

- Monitor API availability.
- Review broken links or missing images.
- Update styling/content periodically.
- Maintain repository backups.
- Review browser compatibility annually.

Future Recommendations

Potential enhancements include:

- Arkansas filming location maps.
- Embedded trailers.
- Accessibility upgrades.

- Expanded database with directors/producers.
- Migration to full database-backed architecture.

SUMMARY/LESSONS LEARNED

Positive Outcomes

- Project completed ahead of schedule.
- Stayed within labor-hour estimates.
- Successfully integrated third-party API services.
- Delivered an engaging and functional proof-of-concept website.
- Demonstrated practical full-stack front-end development skills.

Lessons Learned

- Third-party APIs create dependency risk.
- Reusable templates greatly reduce development time.
- Testing navigation and data edge cases is essential.
- Strong documentation improves project execution.
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LIST OF REFERENCE SOURCES

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APPENDICES

Hours Worked

Project Component	**Component Time (Hours)**	**Actual Time (Hours)**	**Status**	**Team Member(s)**
Concept Development	40	40	Complete	
Code Development	160	160	Complete	
Testing	30	30	Complete	
Deployment	15	15	Complete	
Total	**245 hours**	**245 hours**	**Complete**	

Documentation

- Project Charter
- Project Analysis and Design
- Testing logs
- Final Presentation PowerPoint
- Source Code: https://melissae50.github.io/portfolio/arkansasFilmDatabaseProject/index.html